Tyler Gaines

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Professional Summary

Computer Science student with a strong interest in cybersecurity and a practical, efficiency-first approach to problem-solving. Experienced in designing and benchmarking complex data-driven systems, from custom data structures to inverted indexes. Seeking a cybersecurity internship to apply technical skills in a professional, collaborative environment.

Education

The University of Texas at Dallas Richardson, TX

Bachelor of Science, Computer Science. *Expected May 2028*

Relevant Coursework: Data Structures and Algorithms, Computer Architecture, Embedded Systems

Certifications

- CompTIA Security+, September 2026

Skills

Operating Systems: Linux, UNIX, Windows

Programming Languages: Java, C, C++, Assembly

Tools: GitHub, VS Code, MS Office

Languages: English, Portuguese

Activities & Leadership

Cyber Defense and Response Unit August 2025 – May 2026

Member

Computer Security Group August 2024 – May 2025

Member

Experience

Ride Attendant / Lifeguard June 2026 – August 2026

Schlitterbahn *Galveston, TX*

- Operated water attractions efficiently while maintaining strict safety procedures and ensuring a positive guest experience.
- Assisted guests with questions, directions, ride procedures, and general park needs.
- Monitored pools and ride areas for signs of distress, injury, or drowning and responded quickly to emergencies.
- Performed water rescues and provided first aid when necessary, prioritizing guest safety.

Projects

In-Memory Document Search Engine January 2026 – May 2026

Java

- Designed a multi-index document store with a custom circular buffer, TreeMap, and hash-based inverted index, enabling O(1) average-case term lookup and O(log n) prefix/range queries.
- Implemented multi-term AND search using a size-ordered set-intersection strategy to minimize computation on large posting lists.
- Benchmarked performance across 10K–100K document datasets, achieving ~1.5µs average term search latency and validating theoretical complexity against measured results.
- Maintained consistency across four synchronized data structures through all insert/delete operations.

Gaming PC Build & Configuration May 2024 – May 2024

- Assembled a custom gaming PC, including hardware assembly, Windows installation, driver configuration, and system troubleshooting.
- Configured software and system settings to ensure stable performance.